M5A-R4 / ONE INTEGRATED TRUE-AM CANDIDATE / NOT RELEASED
ST PD54008L-E output network and elliptic LPF traced; PHA-1 interstage adaptation unqualified.
2S BOOST9Vfixed OR4S BUCK9Vfull/7.5Vreduced;exclusive populations. Training3.3V only.
No obsolete multiplier chain or adjustable RF coils. Full RF/AM/thermal acceptance NOT_RUN.
OPEN reservoir/coil/connector definitions and changed interstage match; NO MANUFACTURING.
DF782 and TI RNW manufacturer-derived candidate lands checked; thermal/stencil review OPEN.
See docs/M5A_R4_REVIEW_DECISION.md.

01 Interface and local SI5351 source

File: 01_interface_source.kicad_sch

02 Minimal independent fail-off and captured ARM context

File: 02_interlocks.kicad_sch

02 Minimal independent fail-off and captured ARM context continued

File: 02_interlocksb1.kicad_sch

03 External protected pack and exclusive conversion populations

File: 03_input_power.kicad_sch

03 External protected pack and exclusive conversion populations continued

File: 03_input_powerb1.kicad_sch

04 2S boost and regulated MMIC rail

File: 04_boost_driver_supply.kicad_sch

05 Sine reconstruction and series AM modulator

File: 05_am.kicad_sch

05 Sine reconstruction and series AM modulator continued

File: 05_amb1.kicad_sch

06 Source filter, economical driver and fixed bias

File: 06_source_driver_bias.kicad_sch

06 Source filter, economical driver and fixed bias continued

File: 06_source_driver_biasb1.kicad_sch

07 Independent very-low-power sine-AM training path

File: 07_training.kicad_sch

07 Independent very-low-power sine-AM training path continued

File: 07_trainingb1.kicad_sch

08 Local voltage, current and thermal trip

File: 08_protection.kicad_sch

08 Local voltage, current and thermal trip continued

File: 08_protectionb1.kicad_sch

06b ST final-stage matching derivative

File: 06b_st_matching.kicad_sch

06c Original ST elliptic harmonic LPF

File: 06c_st_lpf.kicad_sch

10 Local physical switches and clock sequencing

File: 10_enable_detail.kicad_sch

11 External supplies and integrated commissioning

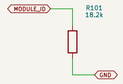
File: 11_external_contract.kicad_sch

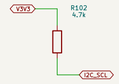
ARDF-TX / NOT RELEASED
Sheet: /
File: RF2M.kicad_sch
Title: ARDF-TX 2 m A2A evaluation schematic
Size: A2 | Date: 2026-10-16 | **Rev: M5A-R4**
KiCad E.D.A. 10.0.6 | Id: 1/19

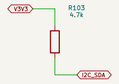
01 Interface and local Si5351 source

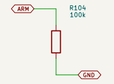
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

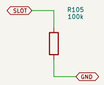


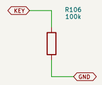


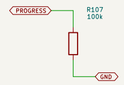


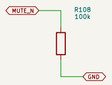


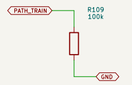


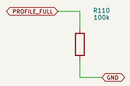


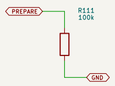


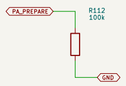


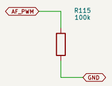


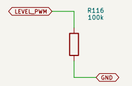




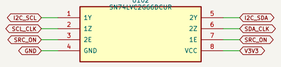


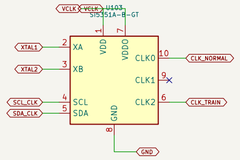


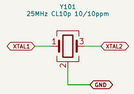


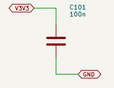


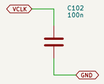


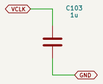
















M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

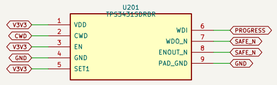
Sheet: /01 Interface and local Si5351 source/
File: 01_Interface_source.kicad_sch

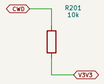
Title: 01 Interface and local Si5351 source

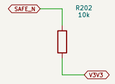
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 2/19

02 Minimal independent fail-off and captured ARM context

M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

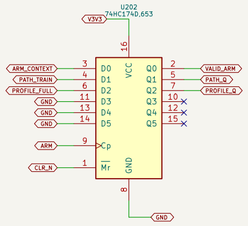














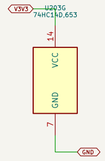


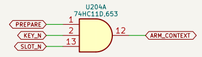




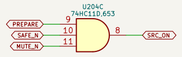


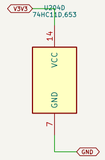


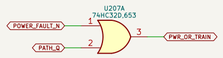




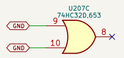












M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context/
File: 02_Interlocks.kicad_sch

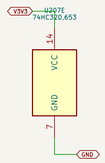
Title: 02 Minimal independent fail-off and captured ARM context

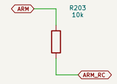
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 3/19

02 Minimal independent fail-off and captured ARM context continued

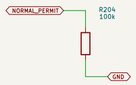
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

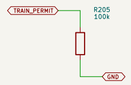




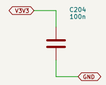


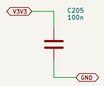


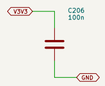


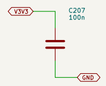


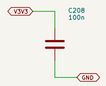


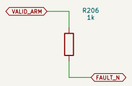


























M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context continued/
File: 02_Interlocksb1.kicad_sch

Title: 02 Minimal independent fail-off and captured ARM context continued

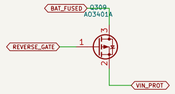
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 4/19

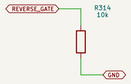
03 External protected pack and exclusive conversion populations

M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.



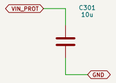


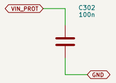


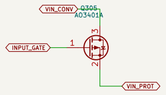


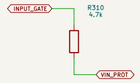


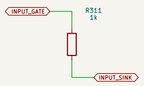


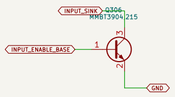


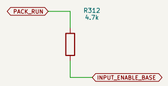


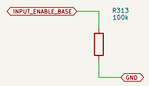






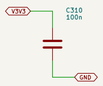




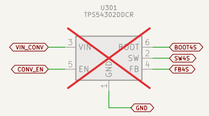




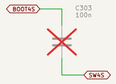


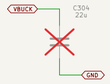




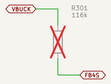






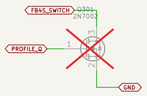


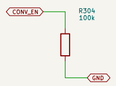


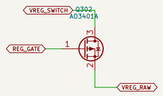


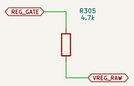












M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

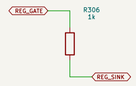
ARDF-TX

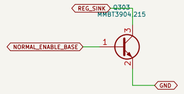
Sheet: /03 External protected pack and exclusive conversion populations/
File: 03_Input_power.kicad_sch

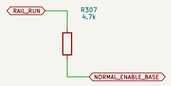
Title: 03 External protected pack and exclusive conversion populations

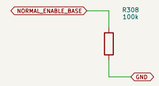
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 5/19

M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

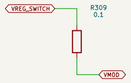






















M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

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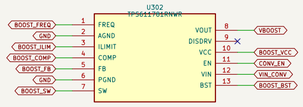
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File: 03_Input_powerb1.kicad_sch

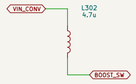
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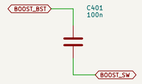
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 6/19

04 2S boost and regulated MMIC rail

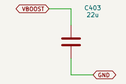
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.



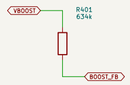


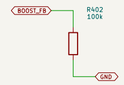


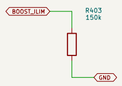


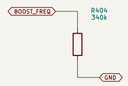


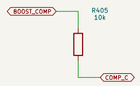


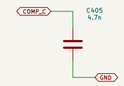


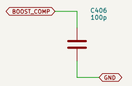


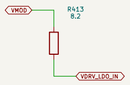


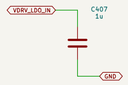




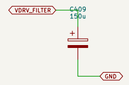
















M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

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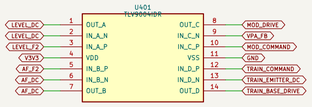
Sheet: /04 2S boost and regulated MMIC rail/
File: 04_boost_driver_supply.kicad_sch

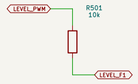
Title: 04 2S boost and regulated MMIC rail

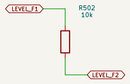
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 7/19

05 Sine reconstruction and series AM modulator

M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

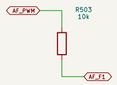


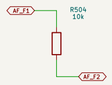






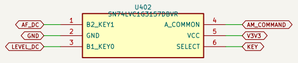


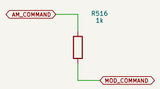


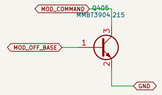


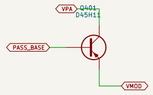


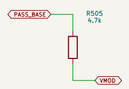


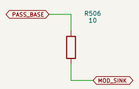


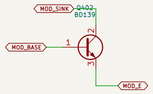


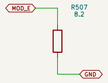


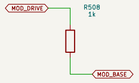


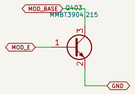


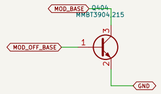


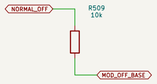


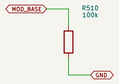


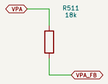


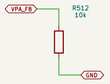


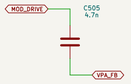




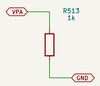




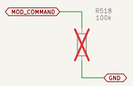


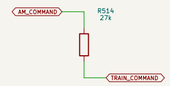












M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

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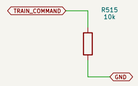
Sheet: /05 Sine reconstruction and series AM modulator/
File: 05_am.kicad_sch

Title: 05 Sine reconstruction and series AM modulator

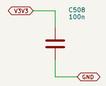
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 8/19

05 Sine reconstruction and series AM modulator continued

M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.













M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

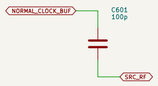
Sheet: /05 Sine reconstruction and series AM modulator continued/
File: 05_amb1.kicad_sch

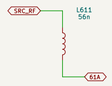
Title: 05 Sine reconstruction and series AM modulator continued

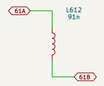
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 9/19

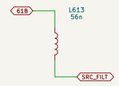
06 Source filter, economical driver and fixed bias

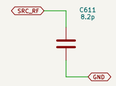
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.



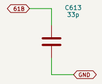


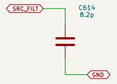


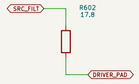


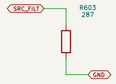


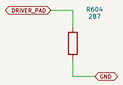






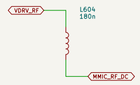


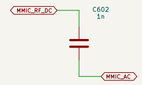


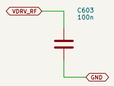




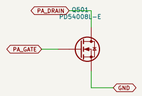


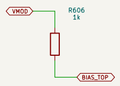


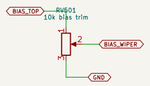




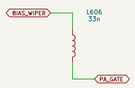


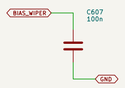


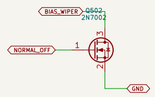














M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

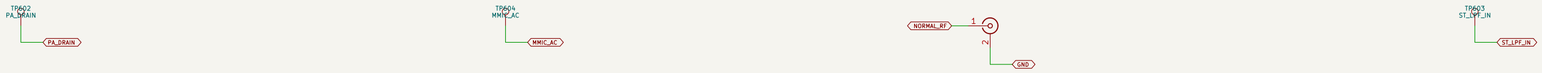
Sheet: /06 Source filter, economical driver and fixed bias/
File: 06_source_driver_bias.kicad_sch

Title: 06 Source filter, economical driver and fixed bias

Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
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06 Source filter, economical driver and fixed bias continued

M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.



M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /06 Source filter, economical driver and fixed bias continued/
File: 06_source_driver_biasb1.kicad_sch

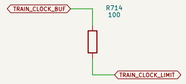
Title: 06 Source filter, economical driver and fixed bias continued

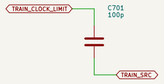
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 11/19

07 Independent very-low-power sine-AM training path

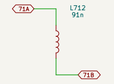
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

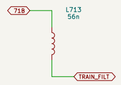


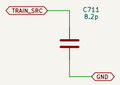


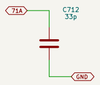


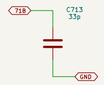




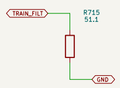


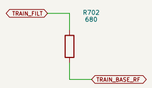


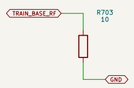


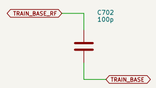


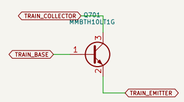


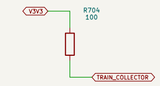


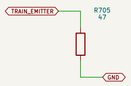


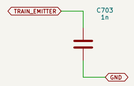


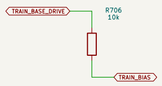


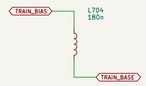


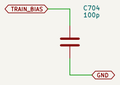


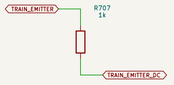


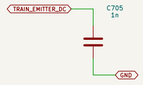


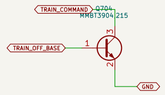


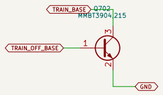


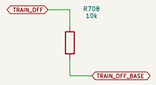


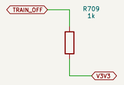


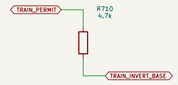


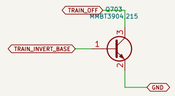


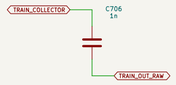












M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

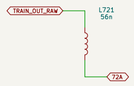
Sheet: /07 Independent very-low-power sine-AM training path/
File: 07_training.kicad_sch

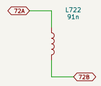
Title: 07 Independent very-low-power sine-AM training path

Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
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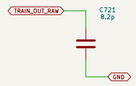
07 Independent very-low-power sine-AM training path continued

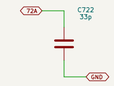
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

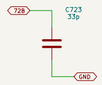


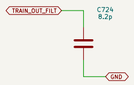


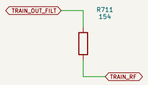


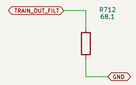


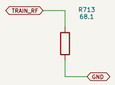


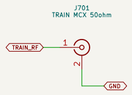
















M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

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Sheet: /07 Independent very-low-power sine-AM training path continued/
File: 07_trainingb1.kicad_sch

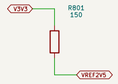
Title: 07 Independent very-low-power sine-AM training path continued

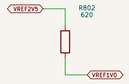
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 13/19

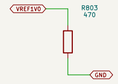
08 Local voltage, current and thermal trip

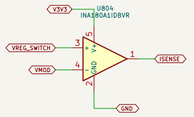
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

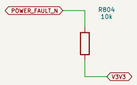


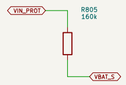


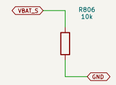


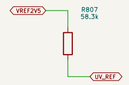


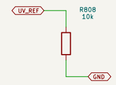


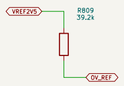


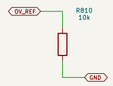


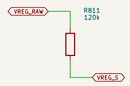


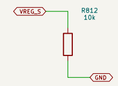


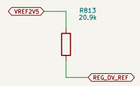


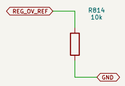


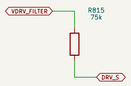


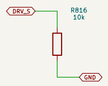


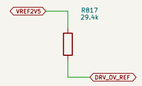


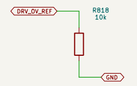


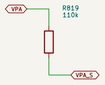


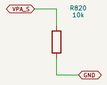


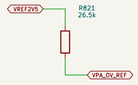


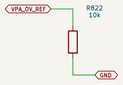


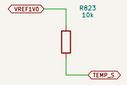


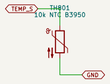


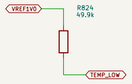


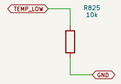


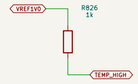


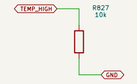












M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

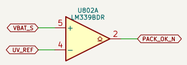
ARDF-TX

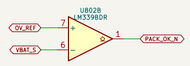
Sheet: /08 Local voltage, current and thermal trip/
File: 08_protection.kicad_sch

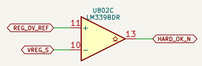
Title: 08 Local voltage, current and thermal trip

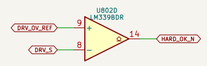
Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 14/19

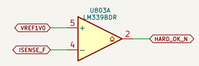
M5A-R4: hardware measurements NOT_RUN: source/reference and simplification in R4 documents.

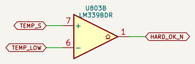


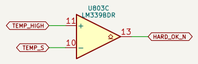


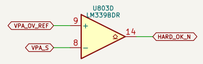


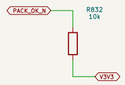


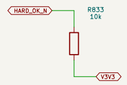










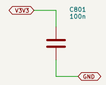


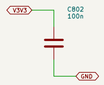


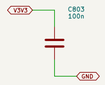


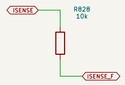


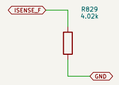


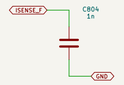






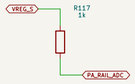


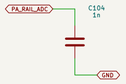


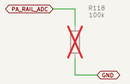












M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

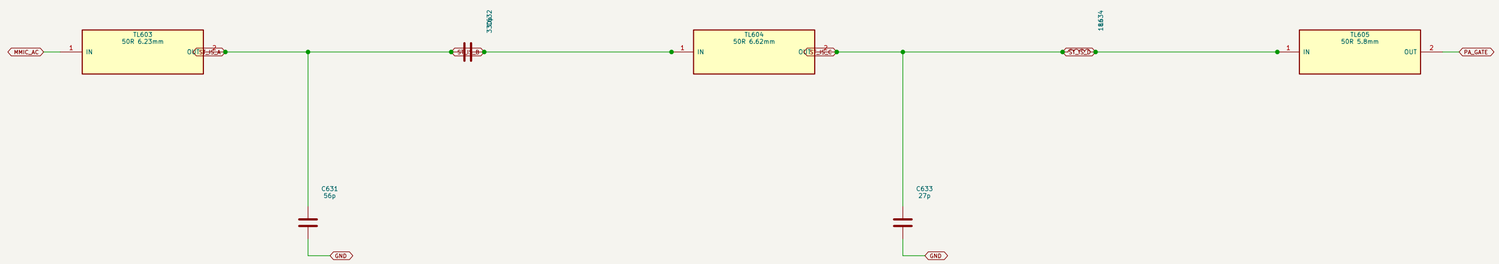
Sheet: /08 Local voltage, current and thermal trip continued/
File: 08_protectionb1.kicad_sch

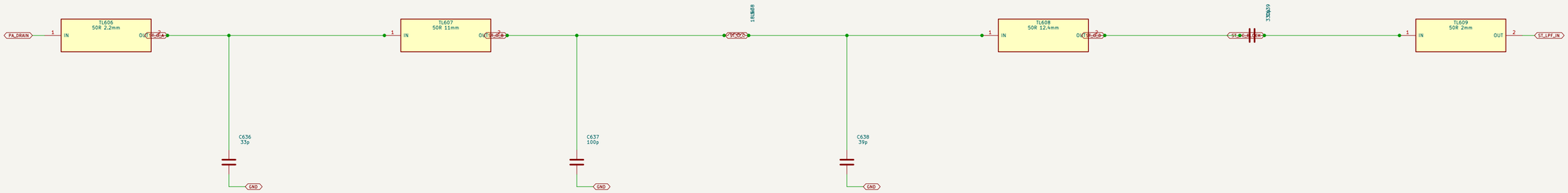
Title: 08 Local voltage, current and thermal trip continued

Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 15/19

06b ST final-stage matching derivative

ST DocID17343 Rev1 Figs16/17/14. Traced topology; changed MMIC drive and supplyAM remain UNQUALIFIED.
TLine symbols are PCB microstrip lengths, NOT purchased components or zero-length wire. No hardware measurements exist.





Input matching: original interstage source was PD84001 drain, not a50ohm MMIC. No claim of unchanged match or sufficient gain.
Output matching/LPF retained. L63533s drain feed is a disclosed substitution for the contradictory35.5n/22n ST BOM entry.
Use one integrated end-to-end commissioning procedure after independent electrical/layout review. No separate PA coupons.

M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

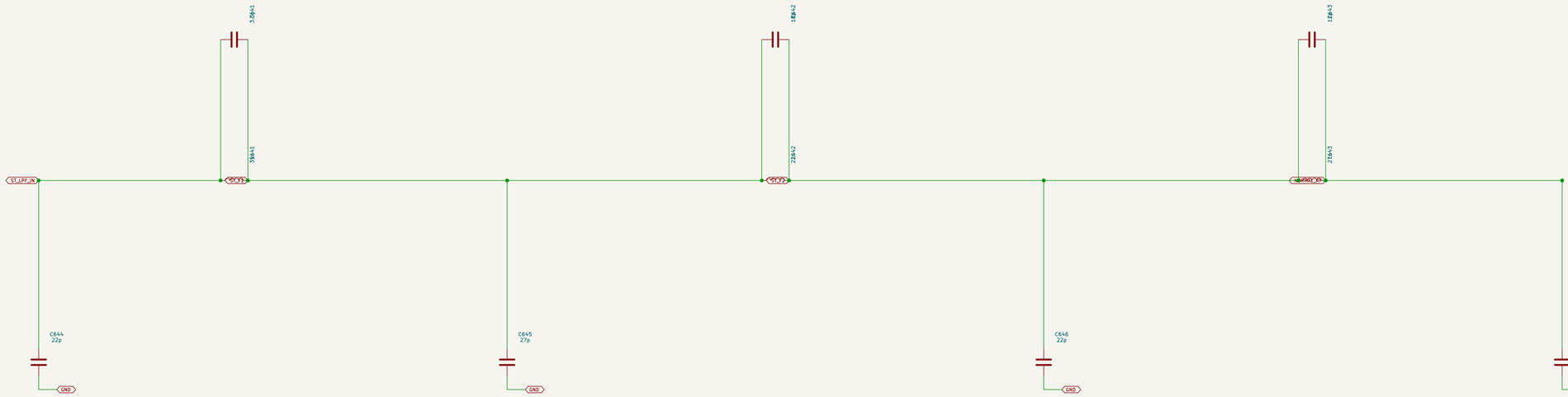
Sheet: /06b ST final-stage matching derivative/
File: 06b_st_matching.kicad_sch

Title: 06b ST final-stage matching derivative

Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 16/19

06c Original ST elliptic harmonic LPF

ST DocID17343 Rev1 Figs16/17/14. Traced topology; changed MMIC drive and supplyAM remain UNQUALIFIED.
TLine symbols are PCB microstrip lengths, NOT purchased components or zero-length wire. No hardware measurements exist.



Input matching: original interstage source was PD84001 drain, not a50ohm MMIC. No claim of unchanged match or sufficient gain.
Output matching/LPF retained. L63533s drain feed is a disclosed substitution for the contradictory35.5n/22n ST BOM entry.
Use one integrated end-to-end commissioning procedure after independent electrical/layout review. No separate PA coupons.

MSA-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

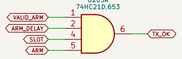
Sheet: /06c Original ST elliptic harmonic LPF/
File: 06c_st_lpf.kicad_sch

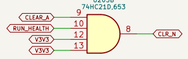
Title: 06c Original ST elliptic harmonic LPF

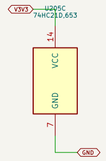
Size: A2	Date: 2026-10-16	**Rev: MSA-R4**
KiCad E.D.A. 10.0.6		Id: 17/19

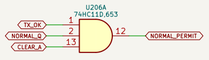
10 Local physical switches and clock sequencing

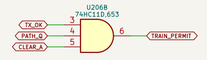
One integrated transmitter, no separate coupon.9V fixed converter, separate filtered5V driver.
TPS22919 replaces discrete driver-switch network. Source gate delays only turn-on; fall-off uses direct inhibit and rail isolation.
RC timings, analog trip latency and real RF silence remain physical acceptance gates.



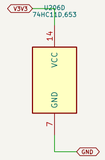




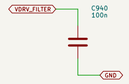


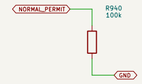


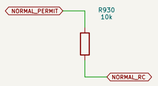


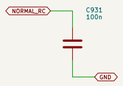


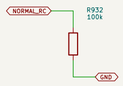




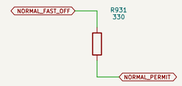


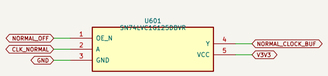


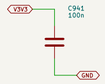














MSA-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /10 Local physical switches and clock sequencing/
File: 10_enable_detail.kicad_sch

Title: 10 Local physical switches and clock sequencing

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11 External supplies and integrated commissioning

R4 MPN/landing blockers are explicit. No PCB manufacturing release. All physical RF/AM/thermal tests NOT_RUN.
The measured antenna gain sets ERP calibration;1dB reserve is optional engineering headroom, NOT an IARU rule.
Controller: R4 ID=144MHz direct clocks,25 fixed9V /4S9Vfull7.5Vreduced;AF/DC calibrated separately; keep20pin pinout.

#FLG950
PWR_FLAG
GND

#FLG951
PWR_FLAG
V3V3

#FLG952
PWR_FLAG
BAT_RAW

#FLG953
PWR_FLAG
VIN_PROT

#FLG954
PWR_FLAG
VIN_CONV

#FLG955
PWR_FLAG
VREG_RAW

#FLG956
PWR_FLAG
VMOD

#FLG957
PWR_FLAG
VBUCK

#FLG958
PWR_FLAG
VBOOST

#FLG959
PWR_FLAG
VDRV_LDO_IN

M5A-R4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /11 External supplies and integrated commissioning/
File: 11_external_contract.kicad_sch

Title: 11 External supplies and integrated commissioning

Size: A2	Date: 2026-10-16	**Rev: M5A-R4**
KiCad E.D.A. 10.0.6		Id: 19/19